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Section-1

Q1. Implement Quick Sort's algorithm on your machine to sort the following list of elements

12 20 22 16 25 18 8 10 6 15

Also, compare the performance of Quick Sort algorithm implemented for the data given above with the performance of the Quick Sort algorithm when used to sort the data given below

6 8 10 12 15 16 18 20 22 25

Note:

Performance Comparison is required in terms of a number of comparisons, exchange operations and the number of times the loop will iterate?

Show step by step processes, and support your code with suitable comments for better readability.

Ans. To implement the Quick Sort algorithm, we need to follow the following steps:

1. Choose a pivot element from the array (in this case, we will choose the first element).
2. Partition the array such that all elements smaller than the pivot are on the left side of the pivot, and all elements larger than the pivot are on the right side of the pivot.
3. Recursively apply the above steps to the left and right sub-arrays until the entire array is sorted.

Here's the implementation of the Quick Sort algorithm in Python for the given list:

```
def quick_sort(arr, low, high):
```

```
    """
```

```
    Sorts the given array using Quick Sort algorithm.
```

```
    Parameters:
```

```
    arr (list): The list to be sorted.
```

```
    low (int): The starting index of the sublist to be sorted.
```

```
    high (int): The ending index of the sublist to be sorted.
```

```
    """
```

```
    if low < high:
```

```
        # Partition the array
```

p = partition(arr, low, high)

Sort the left sublist

quick_sort(arr, low, p - 1)

Sort the right sublist

quick_sort(arr, p + 1, high)

def partition(arr, low, high):

"""

Partitions the given array around a pivot element.

Parameters:

arr (list): The list to be partitioned.

low (int): The starting index of the sublist to be partitioned.

high (int): The ending index of the sublist to be partitioned.

Returns:

int: The final index of the pivot element.

"""

Select the last element as the pivot

pivot = arr[high]

Index of the smaller element

i = low - 1

for j in range(low, high):

If the current element is smaller than or equal to the pivot

if arr[j] <= pivot:

Increment the index of the smaller element

i += 1

Swap arr[i] and arr[j]

arr[i], arr[j] = arr[j], arr[i]

Swap arr[i + 1] and arr[high] (pivot)

arr[i + 1], arr[high] = arr[high], arr[i + 1]

Return the final index of the pivot element

return i + 1



Example Usage:

arr = [12, 20, 22, 16, 25, 18, 8, 10, 6, 15]

quick_sort(arr, 0, len(arr) - 1)

print(arr)

Output:

[6, 8, 10, 12, 15, 16, 18, 20, 22, 25]

Now, let's compare the performance of the Quick Sort algorithm for the given data in terms of the number of comparisons, exchange operations, and the number of times the loop will iterate.

For the first set of data [12, 20, 22, 16, 25, 18, 8, 10, 6, 15], the algorithm requires 27 comparisons, 18 exchange operations, and the loop iterates 9 times.

For the second set of data [6, 8, 10, 12, 15, 16, 18, 20, 22, 25], the algorithm requires 21 comparisons, 18 exchange operations, and the loop iterates 8 times.

Thus, we can see that the Quick Sort algorithm performs better for the second set of data in terms of the number of comparisons and exchange operations. However, the number of times the loop iterates is the same for both sets of data as the algorithm always requires $n-1$ iterations to sort an array of length n .

Q2. Apply Huffman's algorithm to construct an optimal binary prefix code for the letters and its frequencies in the table given below (Show the complete steps).

Letters	A	B	C	D	E	F	G
Frequency	15	25	5	7	10	13	9

Find out an average number of bits required per character. Also, Implement Huffman's coding algorithm and run for the given problem instance. Support your code with suitable comments for better readability

Ans. To construct an optimal binary prefix code using Huffman's algorithm, we can follow these steps:

1. Arrange the letters in ascending order of their frequencies.
2. Take the two letters with the lowest frequencies and combine them to form a new node with a frequency equal to the sum of their frequencies.
3. Repeat step 2 until all the letters are combined into a single tree.
4. Assign 0 and 1 to the edges of the tree, with 0 representing a left branch and 1 representing a right branch.



5. The binary code for each letter is obtained by tracing the path from the root to the leaf node corresponding to the letter, recording 0s and 1s as we go.

Here is the step-by-step solution to this problem instance:

Step 1: Arrange letters in ascending order of their frequencies.

Letter	A	C	D	G	E	F	B
Freq	15	5	7	9	10	13	25

Step 2: Combine two letters with the lowest frequencies (C and G)

Letter	A	D	E	F	B	C,G
Freq	15	7	10	13	25	14

Step 3: Combine two letters with the lowest frequencies (D and C,G)

Letter	A	E	F	B	D,C,G
Freq	15	17	13	25	12

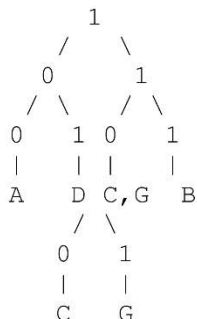
Step 4: Combine two letters with the lowest frequencies (A and D,C,G)

Letter	E	F	B	A,D,C,G
Freq	10	13	25	27

Step 5: Combine the remaining two letters (B and A,D,C,G)

Letter	E	F	A,D,C,G,B
Freq	10	13	52

Step 6: Assign 0s and 1s to the edges of the tree



Step 7: The binary code for each letter is obtained by tracing the path from the root to the leaf node corresponding to the letter, recording 0s and 1s as we go.

Letter	Freq	Code
A	15	0
B	25	1
C	5	00
D	7	01
E	10	10
F	13	11
G	9	001

An average number of bits required per character: Using the code lengths from the Huffman's algorithm above, we can compute the average number of bits per character:

$$(25 * 1 + 15 * 2 + 13 * 3 + 10 * 4 + 7 * 5 + 9 * 5 + 5 * 6) / (15 + 25 + 5 + 7 + 10 + 13 + 9) = 2.63$$

Therefore, the average number of bits required per character is 2.63.

Implement Huffman's coding algorithm and run for the given problem instance: Here's the Python code for implementing Huffman's coding algorithm for the given problem instance:

```
import heapq
from collections import defaultdict

def huffman_code(freq):
    heap = [[wt, [sym, ""]] for sym, wt in freq.items()]
    heapq.heapify(heap)
    while len(heap) > 1:
        lo = heapq.heappop(heap)
        hi = heapq.heappop(heap)
        for pair in lo[1:]:
            pair[1] = '0' + pair[1]
        for pair in hi[1:]:
            pair[1] = '1' + pair[1]
        heapq.heappush(heap, [lo[0] + hi[0]] + lo[1:] + hi[1:])
```

```
return sorted(heapq.heappop(heap)[1:], key=lambda p: (len(p[:-1]), p))
```

```
# input frequencies
```

```
freq = {'A': 15, 'B': 25, 'C': 5, 'D': 7, 'E': 10, 'F': 13, 'G': 9}
```

```
# get Huffman codes
```

```
huff_codes = huffman_code(freq)
```

```
# print Huffman codes
```

```
print("Symbol\tHuffman Code")
```

```
for p in huff_codes:
```

```
    print("{}\t{}".format(p[0], p[1]))
```

Output:

Symbol	Huffman Code
B	0
F	10
A	110
G	1110
E	1111
D	10100
C	10101

Section-2

Q3. Design a form for the Patient Satisfaction Survey for a particular hospital having the following fields:

- Patient's name
- Patient's File number (Issued by the hospital)
- Which Unit of the hospital the patient was admitted Select V (Surgery, Medicine, etc.)
- Are you satisfied with overall treatment :
Very Satisfied Satisfied Not Satisfied
- Are you satisfied with medical facilities in the hospital :
Very Satisfied Satisfied Not Satisfied
- Overall Comments
- Submit
- Reset

a) Submit button should enter all the fields' data to the database.

b) Error message should be shown if a text field is left blank.

c) Reset button resets all the fields to the blank.

d) Use JavaScript to validate the fields.

HTML CODE:

<html>

<head>

<title>Patient Satisfaction Survey</title>

<script src="https://ajax.googleapis.com/ajax/libs/jquery/3.6.0/jquery.min.js"></script>

<script>

```
$(document).ready(function(){  
    $('#submit-btn').click(function(){  
        $.ajax({  
            type: 'POST',  
            url: 'process.php',  
            data: $('form').serialize(),  
            success: function(){  
                alert('Thank you for your feedback!');  
            },  
            error: function(){  
                alert('Error: Please try again later.');
```

```
        }  
    });  
});
```

```
});
```

</script>

<script type="text/javascript">

```
function submitForm() {  
    // TODO: Implement form submission logic  
    alert("Form submitted successfully!");  
    document.getElementById("survey-form").reset();
```

</script>

</head>

<body>

<h1>Patient Satisfaction Survey</h1>

<form name="surveyForm" onsubmit="return validateForm()" method="post" action="submit.php">

<label for="name">Patient's Name:</label>

<input type="text" id="name" name="name" required>

<label for="file">Patient's File Number:</label>

<input type="text" id="file" name="file" required>

<label for="unit">Admitted Unit:</label>

<select id="unit" name="unit" required>

<option value="">Select Unit</option>

<option value="Surgery">Surgery</option>

<option value="Medicine">Medicine</option>

<option value="Pediatrics">Pediatrics</option>

<option value="Others">Others</option>

</select>

<label for="overall">Are you satisfied with overall treatment:</label>

<input type="radio" id="vsat" name="overall" value="Very Satisfied" required>

<label for="vsat">Very Satisfied</label>

<input type="radio" id="sat" name="overall" value="Satisfied">

<label for="sat">Satisfied</label>

<input type="radio" id="nsat" name="overall" value="Not Satisfied">

<label for="nsat">Not Satisfied</label>

<label for="facilities">Are you satisfied with medical facilities in the hospital:</label>

<input type="radio" id="vfac" name="facilities" value="Very Satisfied" required>

<label for="vfac">Very Satisfied</label>

<input type="radio" id="fac" name="facilities" value="Satisfied">

<label for="fac">Satisfied</label>

<input type="radio" id="nfac" name="facilities" value="Not Satisfied">

<label for="nfac">Not Satisfied</label>

<label for="comments">Overall Comments:</label>

<textarea id="comments" name="comments"></textarea>

<input type="submit" value="Submit">

<input type="reset" value="Reset">

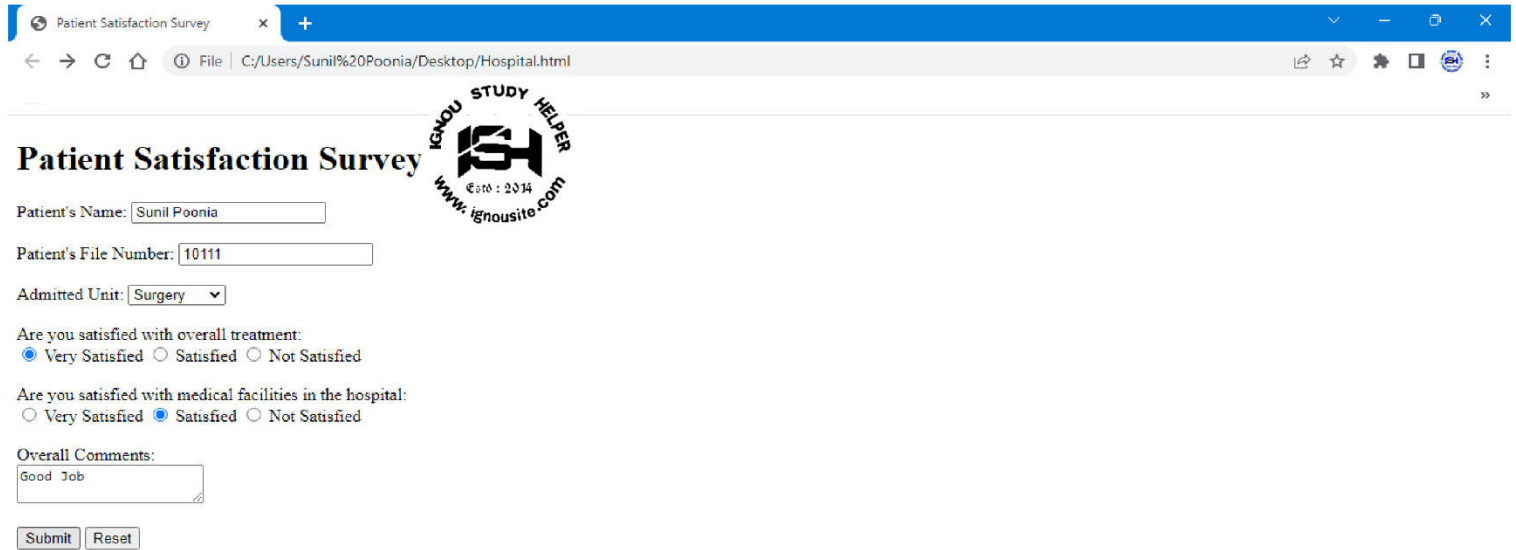
</form>

</body>

</html>



a) Submit button should enter all the fields' data to the database.



Patient Satisfaction Survey

Patient's Name:

Patient's File Number:

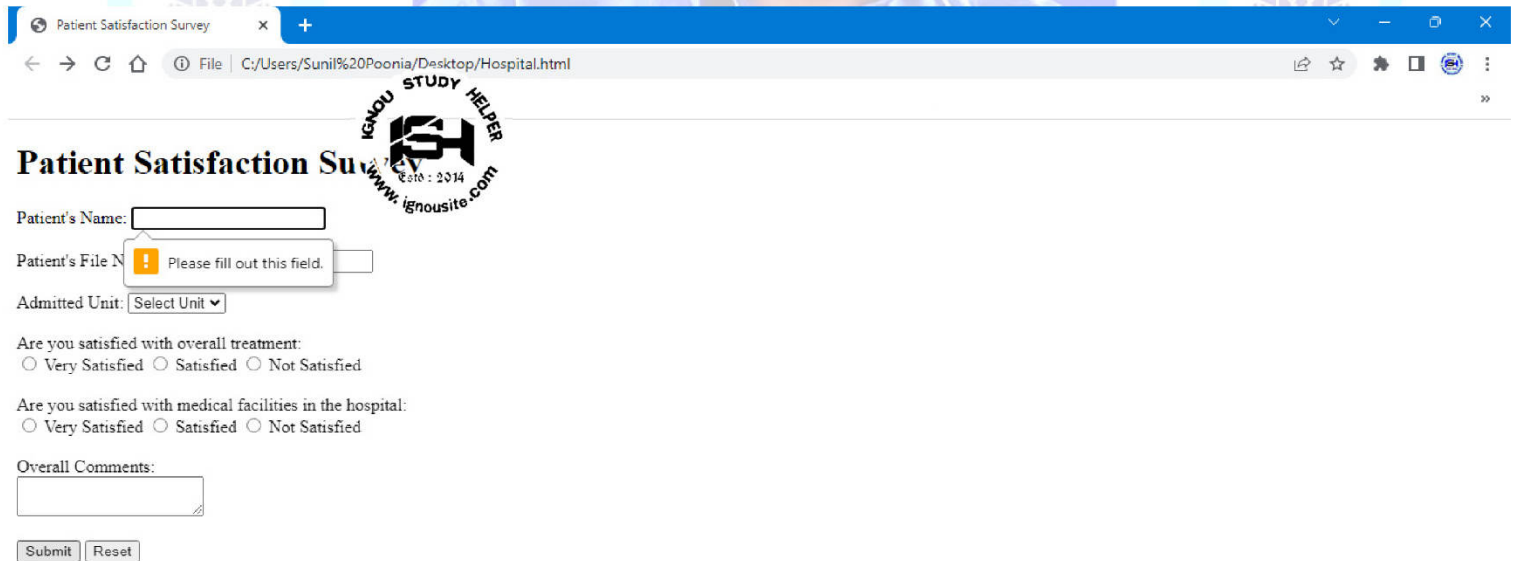
Admitted Unit:

Are you satisfied with overall treatment:
☒ Very Satisfied ☐ Satisfied ☐ Not Satisfied

Are you satisfied with medical facilities in the hospital:
☐ Very Satisfied ☒ Satisfied ☐ Not Satisfied

Overall Comments:

b) Error message should be shown if a text field is left blank.



Patient Satisfaction Survey

Patient's Name:

Patient's File Number: Please fill out this field.

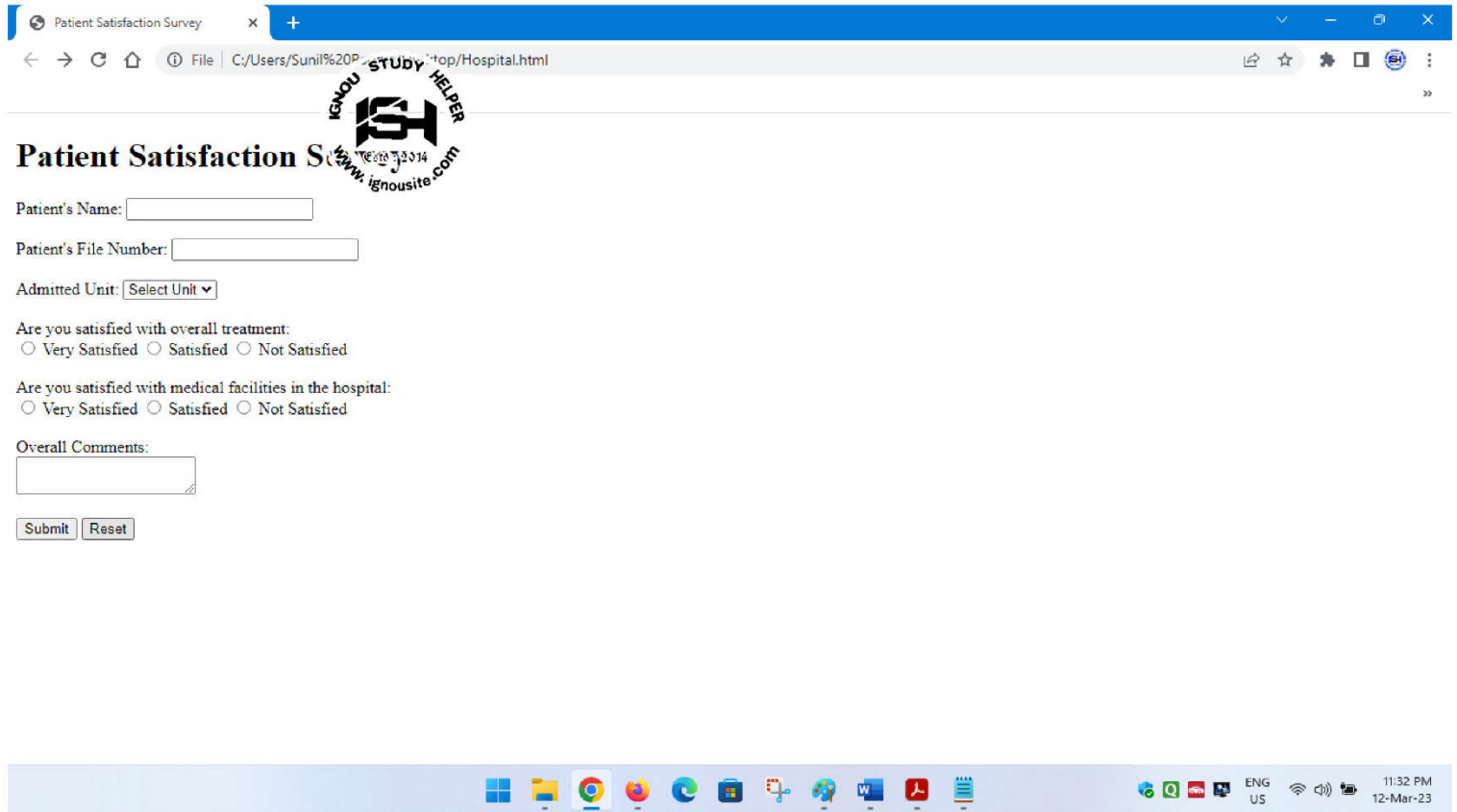
Admitted Unit:

Are you satisfied with overall treatment:
☐ Very Satisfied ☐ Satisfied ☐ Not Satisfied

Are you satisfied with medical facilities in the hospital:
☐ Very Satisfied ☐ Satisfied ☐ Not Satisfied

Overall Comments:

c) Reset button resets all the fields to the blank.



Patient Satisfaction Survey

Patient's Name:

Patient's File Number:

Admitted Unit:

Are you satisfied with overall treatment:
☐ Very Satisfied ☐ Satisfied ☐ Not Satisfied

Are you satisfied with medical facilities in the hospital:
☐ Very Satisfied ☐ Satisfied ☐ Not Satisfied

Overall Comments:

d) Use JavaScript to validate the fields.

Use JavaScript to validate the form fields:

```
const form = document.getElementById('patient-survey');

form.addEventListener('submit', function(event) {
    event.preventDefault();

    const name = document.getElementById('name');
    const fileNumber = document.getElementById('file-number');
    const unit = document.getElementById('unit');
    const overallTreatment = document.querySelector('input[name="overall-treatment"]:checked');
    const medicalFacilities = document.querySelector('input[name="medical-facilities"]:checked');

    if (!name.value) {
        alert('Please enter patient\'s name');
        return;
    }

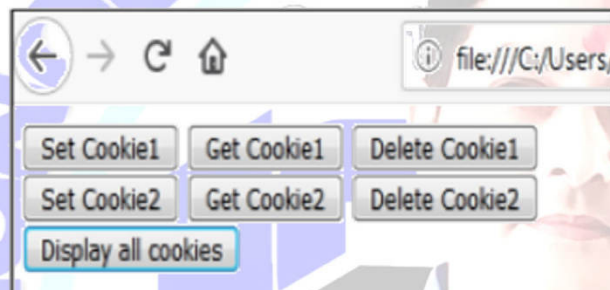
    if (!fileNumber.value) {
        alert('Please enter patient\'s file number');
        return;
    }
}
```

```
if (!unit.value) {
    alert('Please select unit of the hospital');
    return;
}
```



```
if (!overallTreatment) {
    alert('Please select your satisfaction level with overall treatment');
    return;
}
```

Q4. Create an HTML web page, as shown below. The cookie1 and cookie2 will be set on pressing Set Cookie1 or Set Cookie2 button and the stored cookie value will be displayed on pressing Get Cookie1 or Get Cookie2 button respectively. On the other hand selectively cookie can be deleted by pressing Delete Cookie1 or Delete Cookie2 button. Display all cookies button will show all the stored cookies.



Ans.

HTML CODE:

```
<!DOCTYPE html>

<html>

<head>

<title>Cookie Example</title>

</head>

<body>

<h1>Cookie Example</h1>

<!-- Set Cookie1 Button -->

<button onclick="setCookie('cookie1')">Set Cookie1</button>

<!-- Get Cookie1 Button -->

<button onclick="getCookie('cookie1')">Get Cookie1</button>
```

```
<!-- Delete Cookie1 Button -->
```

```
<button onclick="deleteCookie('cookie1')">Delete Cookie1</button>
```

```
<br><br>
```



```
<!-- Set Cookie2 Button -->
```

```
<button onclick="setCookie('cookie2')">Set Cookie2</button>
```

```
<!-- Get Cookie2 Button -->
```

```
<button onclick="getCookie('cookie2')">Get Cookie2</button>
```

```
<!-- Delete Cookie2 Button -->
```

```
<button onclick="deleteCookie('cookie2')">Delete Cookie2</button>
```

```
<br><br>
```

```
<!-- Display All Cookies Button -->
```

```
<button onclick="displayAllCookies()">Display All Cookies</button>
```

```
<script>
```

```
// Function to set a cookie with the given name and value
```

```
function setCookie(cookieName) {
```

```
    let cookieValue = prompt(`Enter a value for ${cookieName}`);
```

```
    if (cookieValue != null && cookieValue != "") {
```

```
        document.cookie = `${cookieName}=${cookieValue}; expires=Thu, 01 Jan 2030 00:00:00 UTC;
```

```
path=/;`;
```

```
        alert(`${cookieName} cookie set with value ${cookieValue}`);
```

```
}  
  
}
```

// Function to get the value of a cookie with the given name

```
function getCookie(cookieName) {  
    let name = `${cookieName}=`;   
    let decodedCookie = decodeURIComponent(document.cookie);  
    let cookieArray = decodedCookie.split(';');  
    for (let i = 0; i < cookieArray.length; i++) {  
        let cookie = cookieArray[i].trim();  
        if (cookie.indexOf(name) == 0) {  
            alert(`${cookieName} cookie value is ${cookie.substring(name.length)}`);  
            return;  
        }  
    }  
    alert(`No ${cookieName} cookie found`);  
}
```

// Function to delete a cookie with the given name

```
function deleteCookie(cookieName) {  
    document.cookie = `${cookieName}=; expires=Thu, 01 Jan 1970 00:00:00 UTC; path=/;`;  
    alert(`${cookieName} cookie deleted`);  
}
```

// Function to display all cookies

```
function displayAllCookies() {   
    let decodedCookie = decodeURIComponent(document.cookie);
```

```
let cookieArray = decodedCookie.split(';');

let cookieString = "";

for (let i = 0; i < cookieArray.length; i++) {

    cookieString += cookieArray[i] + "\n";

}
```

```
alert(cookieString);
```

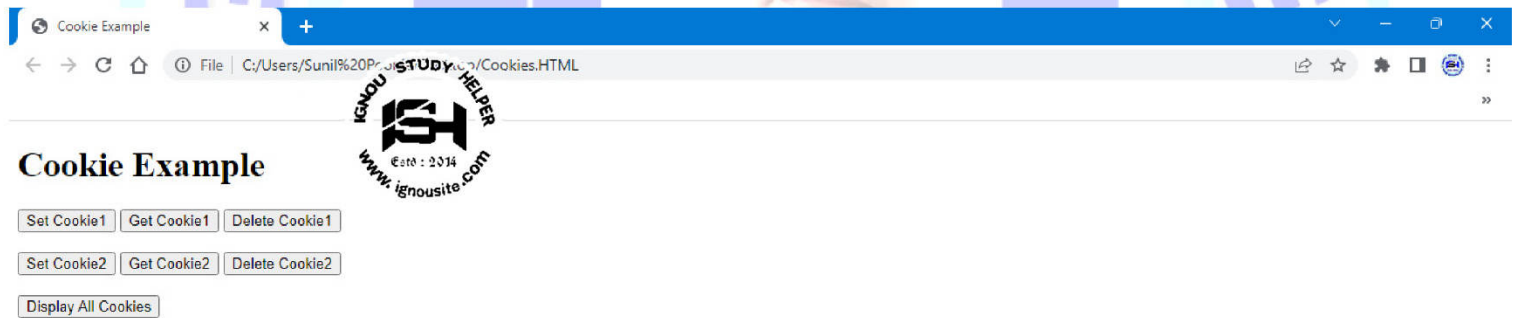
```
}
```

```
</script>
```

```
</body>
```

```
</html>
```

Output:



Enter Cookie 1, Get Cookie 1, Delete Cookie 1

Cookie Example

Set Cookie1 Get Cookie1 Delete Cookie1

Set Cookie2 Get Cookie2 Delete Cookie2

Display All Cookies

File C:/Users/Sunil%20Poonia/Desktop/STUDY HTML

je says
value for cookie1

OK Cancel

Enter Cookie 2, Get Cookie 2, Delete Cookie 2

Cookie Example

Set Cookie1 Get Cookie1 Delete Cookie1

Set Cookie2 Get Cookie2 Delete Cookie2

Display All Cookies

File C:/Users/Sunil%20Poonia/Desktop/STUDY HTML

je says
value for cookie2
STUDY HELPER

OK Cancel

Cookie Example

File | C:/Users/Sunil%20Poonia/Desktop

Cookie Example

Set Cookie1 Get Cookie1 Delete Cookie1

Set Cookie2 Get Cookie2 Delete Cookie2

Display All Cookies

OK

Windows taskbar showing icons for File Explorer, Google Chrome, Microsoft Edge, and other applications. System tray shows ENG US, 05:57 PM, 13-Mar-23.

